

Practical Exam in Analytics Application

Data Wrangling and Visualization

▶ Run Code



```
1 glimpse(branch_data)
2 head(branch_data)
```

Rows: 50

Columns: 7

```
$ Branch      <chr> "Branch 1", "Branch 2", "Branch 3", "Branch 4", "Bra...
$ Zone        <chr> "East", "East", "East", "South", "East", "South", "S...
$ Reported_Date <date> 2023-11-05, 2023-05-15, 2023-11-05, 2023-08-12, 202...
$ Revenue     <dbl> 137823, 110677, 147184, 97973, 96158, 82965, 105423,...
$ COGS        <dbl> 32382, 27971, 38899, 25515, 45909, 42397, 40049, 385...
$ Marketing_Costs <dbl> 9105, 9594, 6413, 9806, 8642, 8432, 5264, 6976, 7389...
$ Operational_Costs <dbl> 10400, 20056, 27421, 10126, 11441, 13284, 25407, 247...
```

	Branch	Zone	Reported_Date	Revenue	COGS	Marketing_Costs	Operational_Costs
1	Branch 1	East	2023-11-05	137823	32382	9105	10400
2	Branch 2	East	2023-05-15	110677	27971	9594	20056
3	Branch 3	East	2023-11-05	147184	38899	6413	27421
4	Branch 4	South	2023-08-12	97973	25515	9806	10126
5	Branch 5	East	2023-06-15	96158	45909	8642	11441
6	Branch 6	South	2023-08-05	82965	42397	8432	13284

Challenge 1

Determine Total Expenses by each Zone.

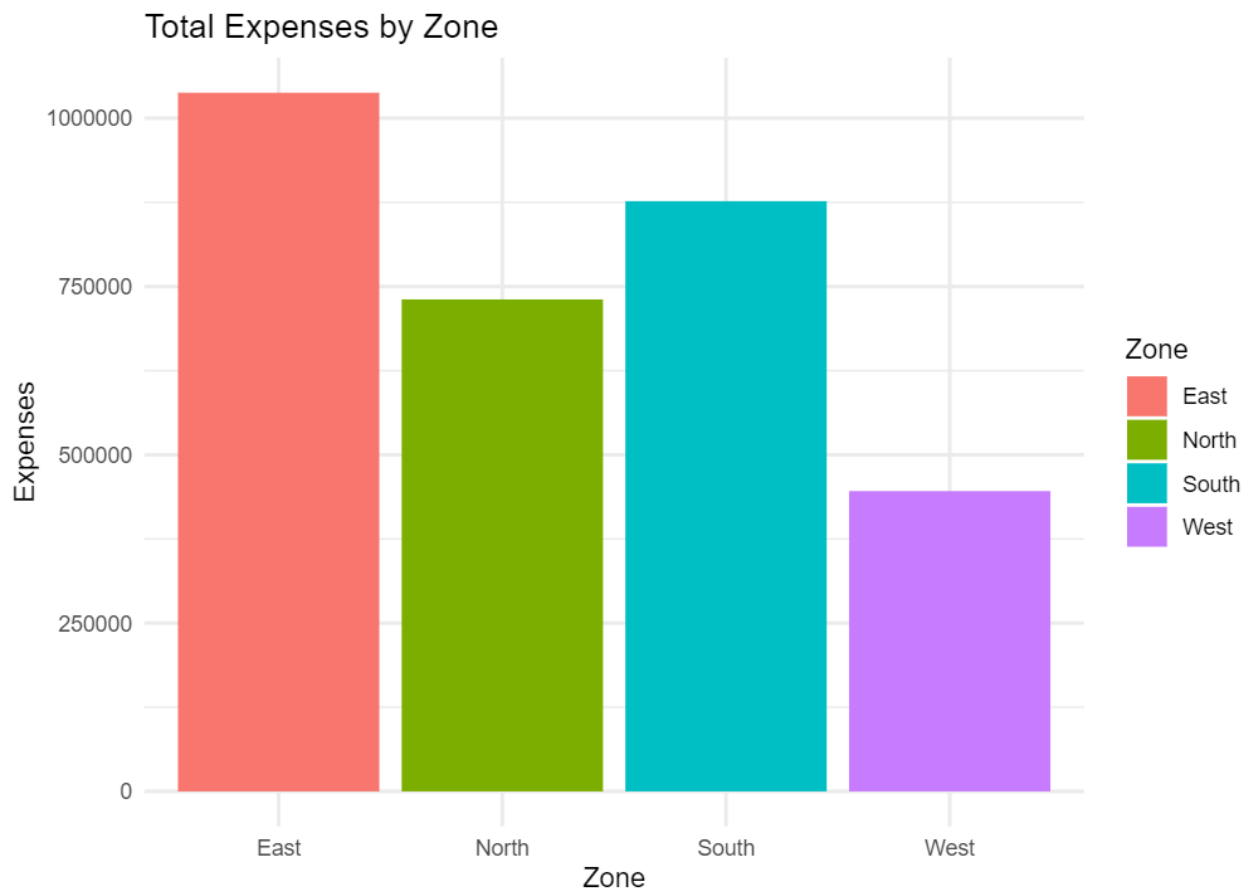
▶ Run Code



```
1 data <- as.data.frame(branch_data)
2 tb_data <- as_tibble(data)
3
4 zone_expenses_data <- tb_data %>% group_by(Zone) %>% summarize
  (Total_Expenses = sum(COGS + Marketing_Costs + Operational_Costs))
5
6 print(head(zone_expenses_data))
7
8 ggplot(zone_expenses_data, aes(x = Zone, y = Total_Expenses, fill = Zone))
  +
9 geom_bar(stat = "identity") +
10 labs(title = "Total Expenses by Zone",
11 x = "Zone",
12 y = "Expenses",
13 fill = "Zone") +
14 theme_minimal()
```

```
# A tibble: 4 × 2
```

	Zone	Total_Expenses
	<chr>	<dbl>
1	East	1037832
2	North	730776
3	South	876734
4	West	446108



Challenge 2

Generate Expenses Plot by Zone and Expense Type. Requires `pivot_longer()`.

Run Code



```
1 data <- as.data.frame(branch_data)
2 tb_data <- as_tibble(data)
3
4 pivot_longer <- tb_data %>% pivot_longer(cols = c(COGS, Marketing_Costs,
5   Operational_Costs),
6   names_to = "Expense_Category",
7   values_to = "Total_Expenses")
8
9 expenses_data <- pivot_longer %>% group_by(Zone) %>% select
  (Expense_Category, Total_Expenses)
```

```

10 print(head(expenses_data))
11
12 ggplot(expenses_data, aes(x = Zone, y = Total_Expenses, fill =
  Expense_Category)) +
13 geom_bar(stat = "identity")+
14 labs(title = "Total Expenses by Zone",
15 x = "Zone",
16 y = "Expenses",
17 fill = "Expense_Category") +
18 theme_minimal()
19

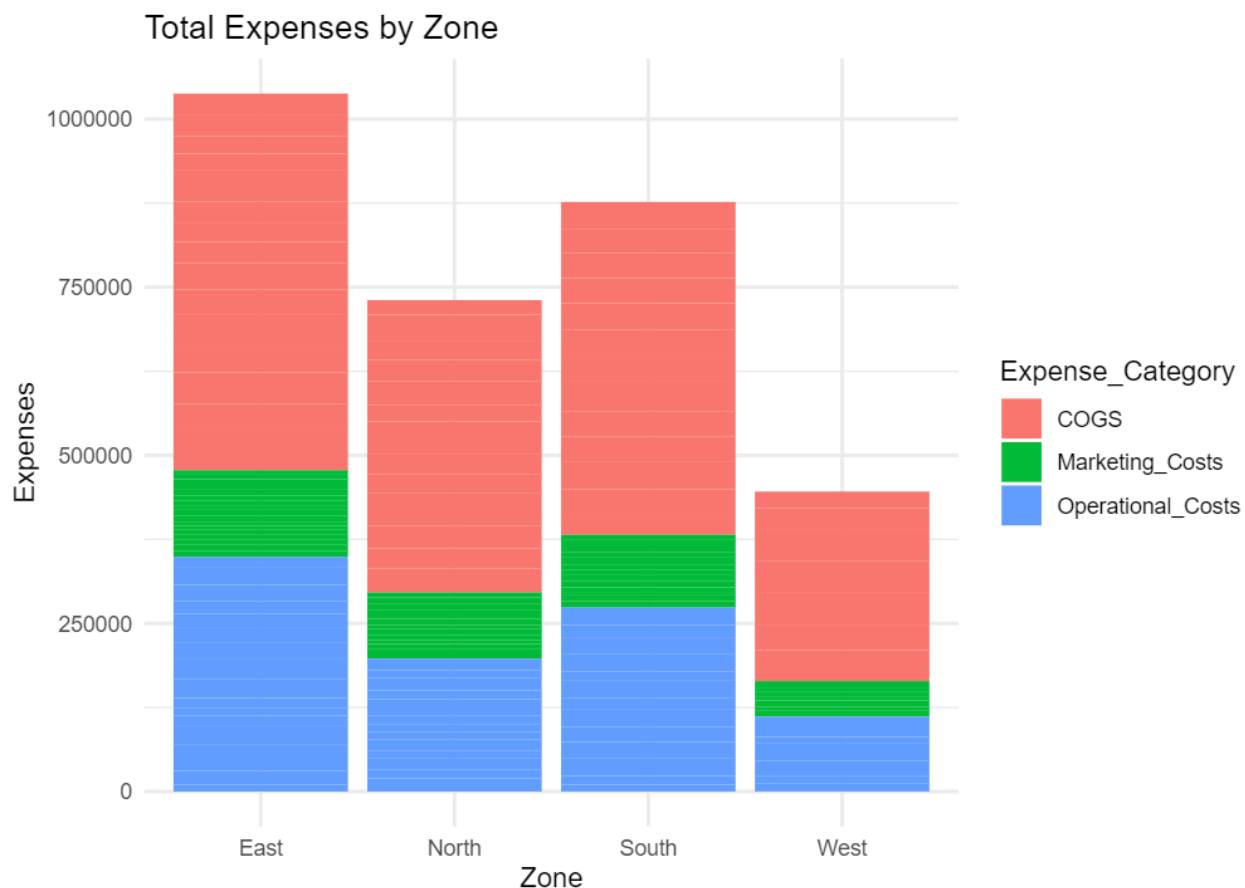
```

Adding missing grouping variables: `Zone`

A tibble: 6 × 3

Groups: Zone [1]

	Zone	Expense_Category	Total_Expenses
	<chr>	<chr>	<dbl>
1	East	COGS	32382
2	East	Marketing_Costs	9105
3	East	Operational_Costs	10400
4	East	COGS	27971
5	East	Marketing_Costs	9594
6	East	Operational_Costs	20056



Challenge 3

Determine Total Profit for each Zone.

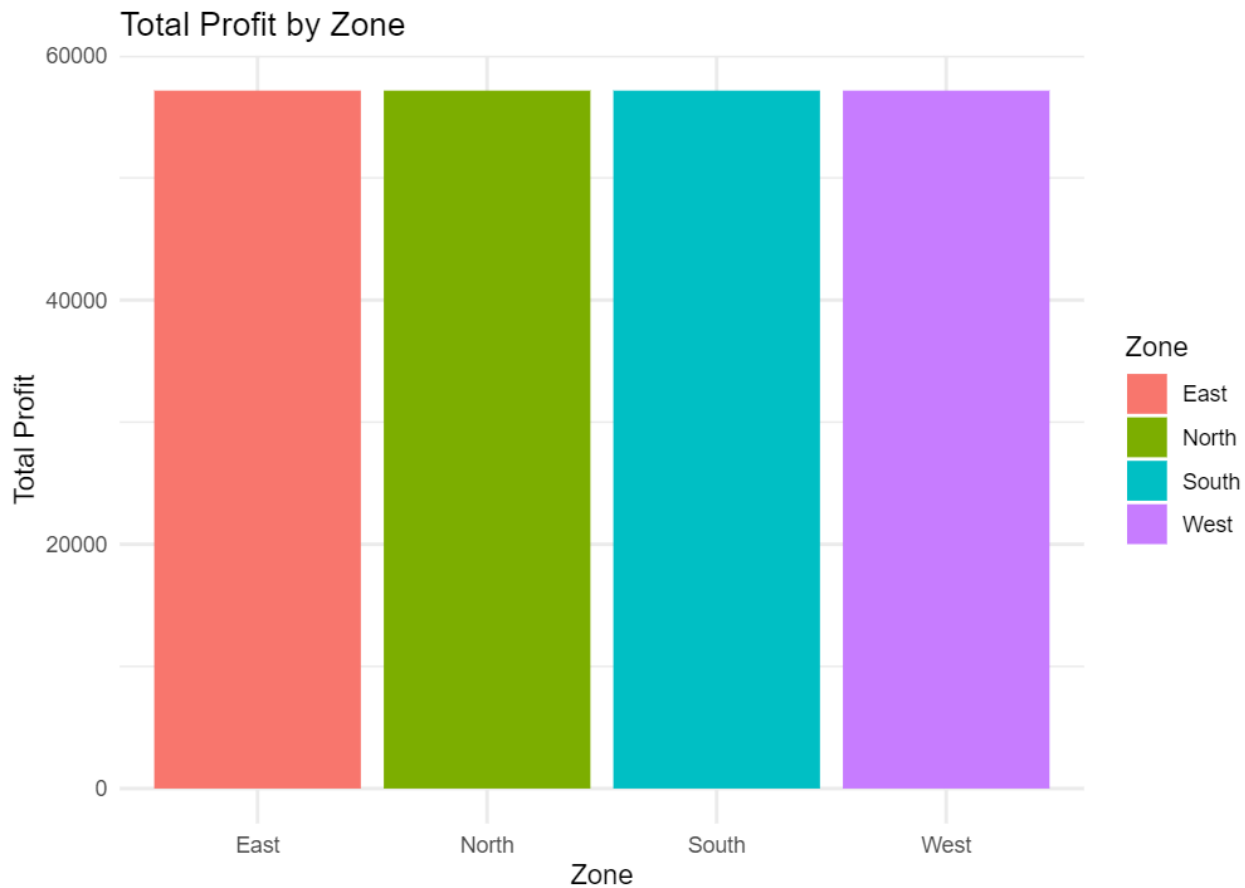
▶ Run Code



```
1 data <- as.data.frame(branch_data)
2 tb_data <- as_tibble(data)
3
4 profit <- tb_data$Revenue - c(tb_data$COGS + tb_data$Marketing_Costs +
  tb_data$Operational_Costs)
5
6 profit_data <- tb_data %>% group_by(Zone) %>% summarize(Total_Profit = mean
  (profit))
7
8 print(head(profit_data))
9
10 ggplot(profit_data, aes(x = Zone, y = Total_Profit, fill = Zone)) +
11   geom_bar(stat = "identity") +
12   labs(title = "Total Profit by Zone",
13    x = "Zone",
14    y = "Total Profit",
15    fill = "Zone") +
16   theme_minimal()
```

A tibble: 4 × 2

	Zone	Total_Profit
	<chr>	<dbl>
1	East	57146.
2	North	57146.
3	South	57146.
4	West	57146.



Challenge 4

Determine Monthly Profit.

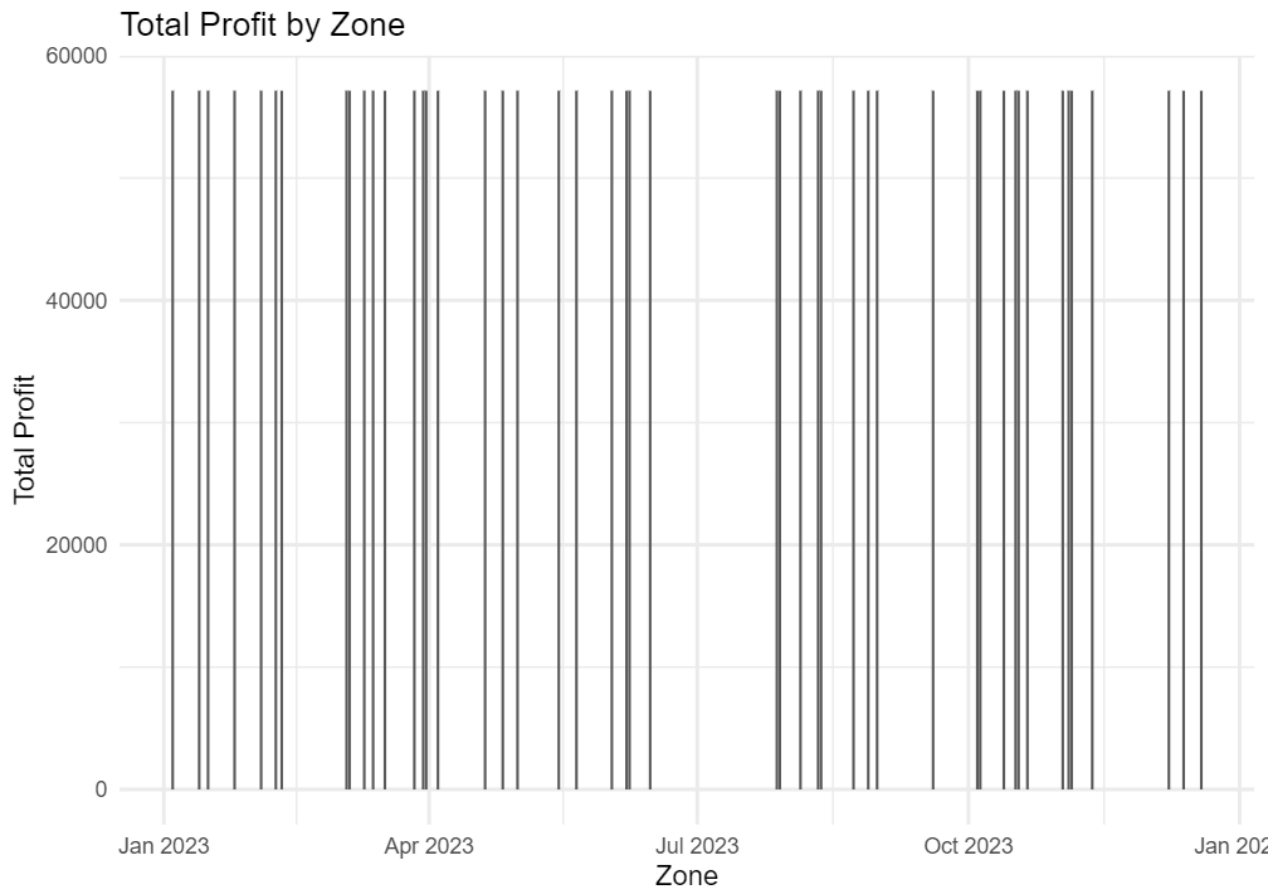
▶ Run Code



```
1 data <- as.data.frame(branch_data)
2 tb_data <- as_tibble(data)
3
4 profit <- tb_data$Revenue - (tb_data$COGS + tb_data$Marketing_Costs +
5   tb_data$Operational_Costs)
6
7 profit_data <- tb_data %>% group_by(Reported_Date) %>% summarize
8   (Total_Profit = mean(profit))
9
10 print(head(profit_data))
11
12 ggplot(profit_data, aes(x = Reported_Date, y = Total_Profit)) +
13   geom_bar(stat = "identity") +
14   labs(title = "Total Profit by Zone",
15     x = "Zone",
16     y = "Total Profit",
17     fill = "Zone") +
18   theme_minimal()
```

A tibble: 6 × 2

	Reported_Date	Total_Profit
	<date>	<dbl>
1	2023-01-04	57146.
2	2023-01-13	57146.
3	2023-01-16	57146.
4	2023-01-25	57146.
5	2023-02-03	57146.
6	2023-02-08	57146.



Challenge 5

Determine Monthly Profit and generate plot faceted by Zone.

Run Code

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```
1 profit <- tb_data$Revenue - (tb_data$COGS + tb_data$Marketing_Costs +
2   tb_data$Operational_Costs)
3 profit_data <- tb_data %>% group_by(Reported_Date, Zone) %>% summarize
4   (Total_Profit = mean(profit))
5 print(head(profit_data))
6
7 ggplot(profit_data, aes(x = Reported_Date, y = Total_Profit)) +
```

```

8 geom_bar(stat = "identity") +
9 facet_wrap(~ Zone)
10 labs(title = "Total Profit by Zone",
11 x = "Zone",
12 y = "Total Profit",
13 fill = "Zone") +
14 theme_minimal()

```

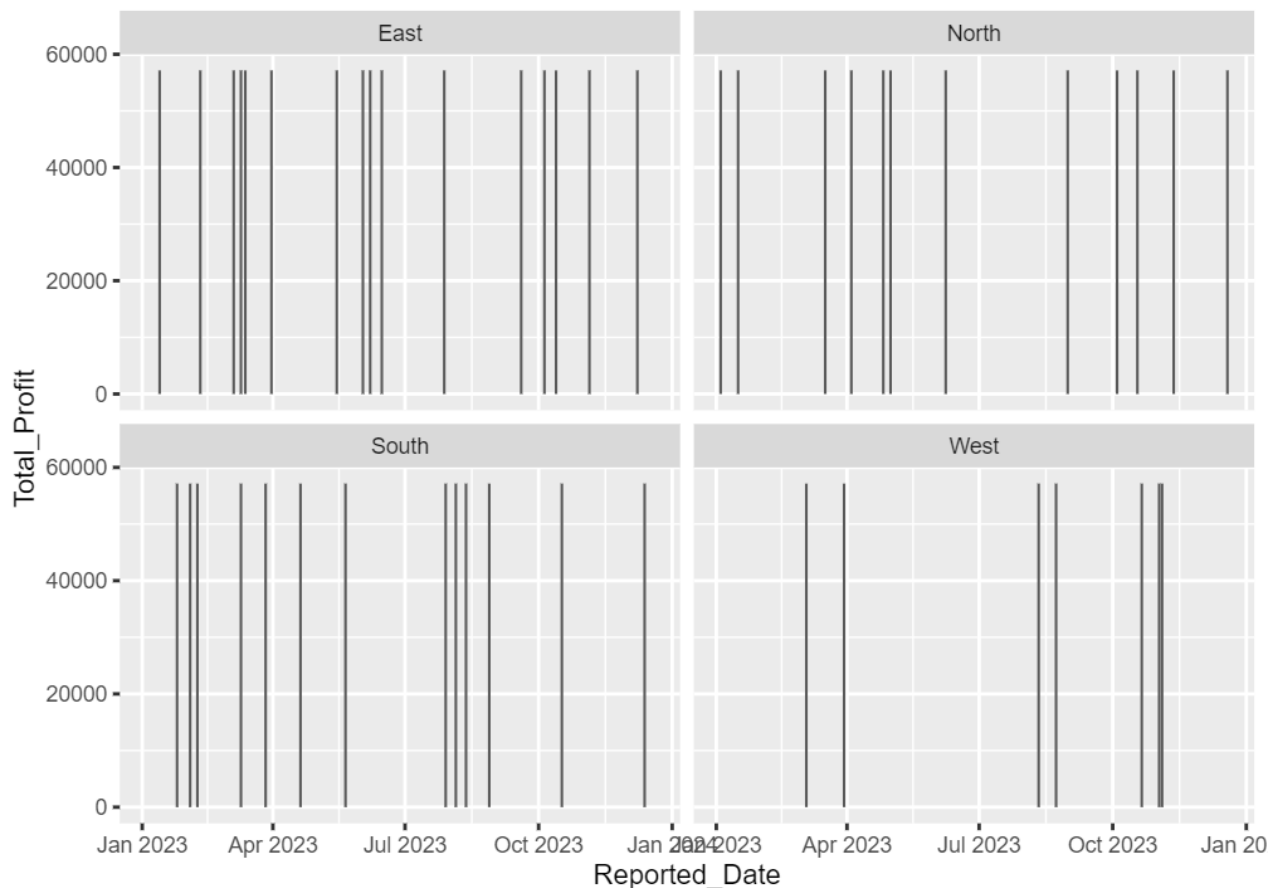
``summarise()`` has grouped output by `'Reported_Date'`. You can override using the ``.groups`` argument.

A tibble: 6 × 3

Groups: Reported_Date [6]

	Reported_Date	Zone	Total_Profit
	<date>	<chr>	<dbl>
1	2023-01-04	North	57146.
2	2023-01-13	East	57146.
3	2023-01-16	North	57146.
4	2023-01-25	South	57146.
5	2023-02-03	South	57146.
6	2023-02-08	South	57146.

NULL



Challenge 6

Determine Total Expense for each Zone and generate plot faceted by Expense Category.

▶ Run Code



```

1  data <- as.data.frame(branch_data)
2  tb_data <- as_tibble(data)
3
4  zone_expenses_data <- tb_data %>% group_by(Zone) %>% summarize
   (Total_Expenses = sum(COGS + Marketing_Costs + Operational_Costs))
5
6  zone_expenses_data <- tb_data %>% pivot_longer(cols = c(COGS,
   Marketing_Costs, Operational_Costs),
7  names_to = "Expense_Category",
8  values_to = "Total_Expenses")
9
10 print(head(zone_expenses_data))
11
12 ggplot(zone_expenses_data, aes(x = Zone, y = Total_Expenses, fill = Zone))
   +
13 geom_bar(stat = "identity") +
14 facet_wrap(~ Expense_Category)
15 labs(title = "Total Expenses by Zone",
16 x = "Zone",
17 y = "Expenses",
18 fill = "Zone") +
19 theme_minimal()

```

A tibble: 6 × 6

	Branch	Zone	Reported_Date	Revenue	Expense_Category	Total_Expenses
	<chr>	<chr>	<date>	<dbl>	<chr>	<dbl>
1	Branch 1	East	2023-11-05	137823	COGS	32382
2	Branch 1	East	2023-11-05	137823	Marketing_Costs	9105
3	Branch 1	East	2023-11-05	137823	Operational_Costs	10400
4	Branch 2	East	2023-05-15	110677	COGS	27971
5	Branch 2	East	2023-05-15	110677	Marketing_Costs	9594
6	Branch 2	East	2023-05-15	110677	Operational_Costs	20056
	NULL					

